Experiment No. 10 – Mini Project

Aim: "To design and develop a Water Supply Android application for managing customer registration, water orders, admin operations, order history, and monthly billing using Android Studio and Firebase."

Apparatus / Software Required :

Hardware	- Laptop(ASUS Vivobook 16x) - Minimum 8 GB RAM - AMD Ryzen 5 5600H with Radeon Graphics Processor - Internet Connection
Software	- Windows 11 Operating System - Android Studio (Quail) - Java Development Kit (JDK) - Android SDK - Android Emulator (Pixel 7) - Gradle Build System

Objectives

- To develop a user-friendly Android application for water supply management.
- To implement customer and admin authentication.
- To provide Normal and Cold water ordering options.
- To manage water orders and order cancellation.
- To implement cash and online payment options.
- To maintain order history and monthly package records.
- To generate monthly bills based on delivered orders.
- To store user and order information using Firebase.
- To implement profile management and logout functionality.
- To test and debug the application.

Theory

A mini project combines the concepts learned throughout Android application development into a complete working application. It includes user interface design, authentication, database integration, navigation, data management, and testing.

The Water Supply Android application is designed to simplify the process of ordering drinking water and managing customer orders. It provides separate functionalities for customers and administrators.

Customers can register, log in, select water types, place orders, view order history, and check their current monthly bill. Normal water costs ₹25, while Cold water costs ₹30. Customers can choose cash or online payment for individual orders. The monthly package maintains order records for generating bills at the end of the month.

The application uses Android Studio with Java and XML for application development, Firebase Authentication for user authentication, and Cloud Firestore for storing and managing application data. The administrator manages orders and monthly billing records, while customers can access their order and billing information.

Main Features

Feature	Technology
Registration and Login	Firebase Authentication
Customer and Admin Modules	Java Activities and XML
Water Ordering	Java and Firebase
Water Type Selection	Normal and Cold Water
Order History	Cloud Firestore
Monthly Packages and Billing	Java and Firebase
Payment Options	Cash and Online Payment
Customer Profile	Firebase and Android UI
Navigation	Activities and Intents

System Working Flow

Customer Registration/Login → Customer Dashboard → Select Water Type → Place Order → Store Order Details in Firebase → Admin Reviews and Updates Order Status → Customer Views Order History → Generate Monthly Bill → View Current Bill and Payment Status.

Procedure

1. Create and configure the Android project in Android Studio.
2. Design the application screens using XML layouts.
3. Implement customer registration and login using Firebase Authentication.
4. Create separate customer and administrator modules.
5. Connect the application to Firebase Cloud Firestore.

6. Implement water selection for Normal (₹25) and Cold (₹30).
7. Develop the order placement and cancellation functionality.
8. Store and retrieve customer order details.
9. Implement order history and order status management.
10. Provide cash and online payment options.
11. Implement monthly package records and bill generation.
12. Develop the customer current-bill screen.
13. Implement profile management and logout.
14. Run the application on an Android device or emulator.
15. Test the authentication, ordering, database, and billing modules.
16. Debug errors and verify the application's functionality.

Testing

The following modules are tested:

- Customer Registration and Login
- Admin Authentication
- Dashboard Navigation
- Water Type Selection
- Order Placement and Cancellation
- Order History and Status
- Firebase Data Storage
- Monthly Package and Bill Generation
- Current Bill Display
- Profile Management and Logout

Expected Outcome

The completed application should provide a convenient platform for customers to order drinking water, select the required water type, view order history, and check monthly bills. It should help administrators manage customer orders and maintain billing records efficiently.

Conclusion

The Water Supply Android application demonstrates the practical implementation of Android application development using Java, XML, Firebase Authentication, and Cloud Firestore. It integrates customer authentication, water ordering, order management, and monthly billing into a single application. The project provides a digital solution for simplifying water supply operations and improving customer service.

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